

Incident-Style Traffic Analysis Report

Lab: Network Traffic Monitoring Lab **Date:** June 2026 **Analyst:** Satwika Gunugu **Tools:** Wireshark, Nmap, nslookup

1. Executive Summary

This report documents traffic captured during a controlled home-lab exercise designed to simulate and analyze common network reconnaissance and credential-exposure behaviors. Three categories of activity were captured and analyzed: an active port scan, plaintext credential transmission over HTTP, and DNS query traffic. Each finding is mapped to its corresponding MITRE ATT&CK technique, with detection and mitigation recommendations provided.

2. Scope and Methodology

- **Capture interface:** Wi-Fi adapter, host machine (192.168.1.3)
- **Target/observed device:** Home router, (192.168.1.1)
- **External test target:** (demo.testfire.net) ((65.61.137.117)), a publicly available, intentionally vulnerable banking-application demo used for security training
- **Capture tool:** Wireshark (saved as (wireshark_Wi-Fi.pcapng))
- **Scan tool:** Nmap 7.99, SYN scan ((-sS))

All activity was generated by the analyst against their own home network equipment and against a publicly sanctioned test application. No third-party production systems were targeted.

3. Finding 1 — Port Scan Activity

Description

An Nmap SYN scan (`nmap -sS 192.168.1.1`) was run against the local router to simulate reconnaissance traffic. The scan completed in approximately 6.6 seconds and returned the following results:

Port	State	Service
23	filtered	telnet
53	open	domain
80	open	http
443	open	https

The router's MAC address resolved to a ZTE OUI, consistent with an ISP-provided device.

Evidence

- `screenshots/01_nmap_scan_results.png` — raw Nmap terminal output
- `screenshots/02_wireshark_port_scan_syn_packets.png` — Wireshark capture filtered on `tcp.flags.syn==1 and tcp.flags.ack==0`, showing the SYN packets generated by the scan

Analysis

The scan generated a burst of TCP SYN packets directed at the router across multiple destination ports in a short time window — the classic signature of port scanning. Port 23 (telnet) being **filtered** rather than open or closed suggests the router actively drops

telnet probes at the firewall level, which is a reasonable hardening measure. Ports 53, 80, and 443 being open is expected, as these support the router's DNS relay and web management interface.

MITRE ATT&CK Mapping

T1046 – Network Service Scanning Adversaries use scanning to discover open ports and running services as a precursor to exploitation.

Recommendations

- Disable remote/WAN-facing management interfaces if not required.
 - Ensure telnet remains disabled or blocked (already observed as filtered).
 - Monitor for repeated SYN bursts across multiple ports from a single source as a scan-detection rule.
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4. Finding 2 — Unencrypted Credential Transmission

Description

A login form was submitted over plain HTTP to `demo.testfire.net/doLogin`. The form data was transmitted as `application/x-www-form-urlencoded`, with no transport encryption (HTTP, not HTTPS).

Evidence

- `screenshots/03_wireshark_http_post_plaintext_credentials.png` — Wireshark "Follow HTTP Stream" view showing the raw POST body, with username and password fields fully readable in plaintext

Analysis

Because the session used HTTP rather than HTTPS, the entire POST body — including

the submitted username and password — was visible in plaintext within the packet capture. Any party able to observe traffic on the same network segment (e.g., via ARP spoofing, a compromised switch/AP, or a man-in-the-middle position) could trivially extract these credentials without needing to break any encryption.

MITRE ATT&CK Mapping

T1040 – Network Sniffing Adversaries sniff network traffic to capture sensitive information such as credentials.

T1557 – Adversary-in-the-Middle Relevant if an attacker actively intercepts traffic between the client and server rather than passively observing it.

Recommendations

- Enforce HTTPS (TLS) for all authentication endpoints; redirect HTTP login attempts to HTTPS.
 - Implement HSTS (HTTP Strict Transport Security) to prevent downgrade to plaintext HTTP.
 - Use network segmentation and 802.1X/WPA2-Enterprise on Wi-Fi to reduce the risk of local sniffing.
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5. Finding 3 — DNS Query Behavior

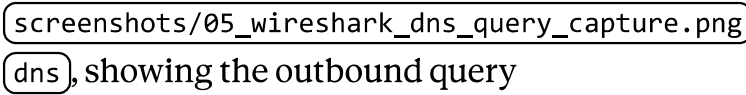
Description

A manual DNS lookup was issued for a non-existent test domain (`something-random-test123.local`) using `nslookup`. The corresponding query was observed in the packet capture.

Evidence

- `screenshots/04_terminal_nslookup_query.png` — terminal output of the nslookup

command

-  — Wireshark capture filtered on `dns`, showing the outbound query

Analysis

DNS queries are sent in plaintext by default (unless DoH/DoT is configured), making them visible to anyone monitoring the network. While this particular query was benign and intentionally generated for the lab, unusual or high-entropy DNS queries in a real environment (e.g., long random subdomains, queries to newly registered domains, or high query volume to a single domain) are a known indicator of DNS tunneling or C2 beaconing.

MITRE ATT&CK Mapping

T1071.004 - Application Layer Protocol: DNS Adversaries may use DNS for command-and-control communication or data exfiltration, blending malicious traffic with legitimate DNS activity.

Recommendations

- Deploy DNS logging/monitoring (e.g., via Zeek, Splunk, or a DNS firewall) to flag anomalous query patterns.
- Consider DNS-layer filtering (e.g., blocking newly observed or algorithmically generated domains).
- Encrypt DNS traffic (DoH/DoT) where supported to reduce passive observability, while ensuring enterprise visibility is preserved through endpoint logging.

6. Overall Recommendations

1. Enforce TLS everywhere — no authentication or sensitive data should ever traverse the network in plaintext.

2. Monitor for scanning behavior at the network perimeter and internally (host-based or network IDS rules for SYN scan patterns).
 3. Add DNS visibility and anomaly detection to the monitoring stack, since DNS is a common and often-overlooked C2/exfiltration channel.
 4. Periodically review router/firewall configurations to confirm unnecessary services (e.g., telnet) remain disabled.
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7. MITRE ATT&CK Summary Table

Technique ID	Technique Name	Observed Behavior
T1046	Network Service Scanning	Nmap SYN scan against router
T1040	Network Sniffing	Plaintext credential capture via packet analysis
T1557	Adversary-in-the-Middle	Risk enabled by lack of HTTPS on login form
T1071.004	Application Layer Protocol: DNS	DNS query observed in cleartext